

EXPERIMENT-11

Q1. Find the sum of the squares of its digits using a while loop.

1. Algorithm

Step 1: Start.

Step 2: Declare integer variables num, choice, n, digit, sumSquares and sumCubes.

Step 3: Initialize sumSquares = 0 and sumCubes = 0.

Step 4: Input a positive integer from the user and store it in num.

Step 5: Check whether num is positive.

If $\text{num} \leq 0$, display "Please enter a positive integer" and stop the program.

Otherwise, continue.

Step 6: Display the menu:

1. Find sum of squares of digits using while loop.

2. Find sum of cubes of digits using for loop.

3. Find difference between sum of cubes and sum of squares using do-while loop.

Step 7: Input the user's choice.

Step 8: Use switch(choice) to perform the selected operation.

Case 1 — Sum of squares using while loop

1. Set $n = \text{num}$.

2. Repeat while $n > 0$:

Find the last digit using $\text{digit} = n \% 10$.

Calculate $\text{sumSquares} = \text{sumSquares} + \text{digit} * \text{digit}$.

Remove the last digit using $n = n / 10$.

3. Display sumSquares.

Case 2 — Sum of cubes using for loop

1. Set $n = \text{num}$.

2. Repeat using a for loop while $n > 0$:

Find the last digit using $\text{digit} = n \% 10$.

Calculate $\text{sumCubes} = \text{sumCubes} + \text{digit} * \text{digit} * \text{digit}$.

Remove the last digit using $n = n / 10$.

3. Display sumCubes.

Case 3 — Difference using do-while loop

1. Set $n = \text{num}$.

2. Using a do-while loop:

Find the last digit using $\text{digit} = n \% 10$.

Add its square to sumSquares.

Add its cube to sumCubes.

Remove the last digit using $n = n / 10$.

3. Continue while $n > 0$.

4. Calculate: $\text{Difference} = \text{sumCubes} - \text{sumSquares}$

5. Display the difference.

Step 9: If the choice is not 1, 2, or 3, display "Invalid choice".

Step 10: Stop.

2. Test Case Table

Test Case	Input (Number, Choice)	Expected Output	Actual Output	Result
1	123, 1	Sum of squares = 14	Sum of squares = 14	Pass
2	123, 2	Sum of cubes = 36	Sum of cubes = 36	Pass
3	123, 3	Difference = 22	Difference = 22	Pass
4	45, 1	Sum of squares = 41	Sum of squares = 41	Pass
5	45, 2	Sum of cubes = 189	Sum of cubes = 189	Pass
6	45, 3	Difference = 148	Difference = 148	Pass

Verification of example

For 123:

Sum of squares:

$$\begin{aligned}1^2 + 2^2 + 3^2 \\&= 1 + 4 + 9 \\&= 14\end{aligned}$$

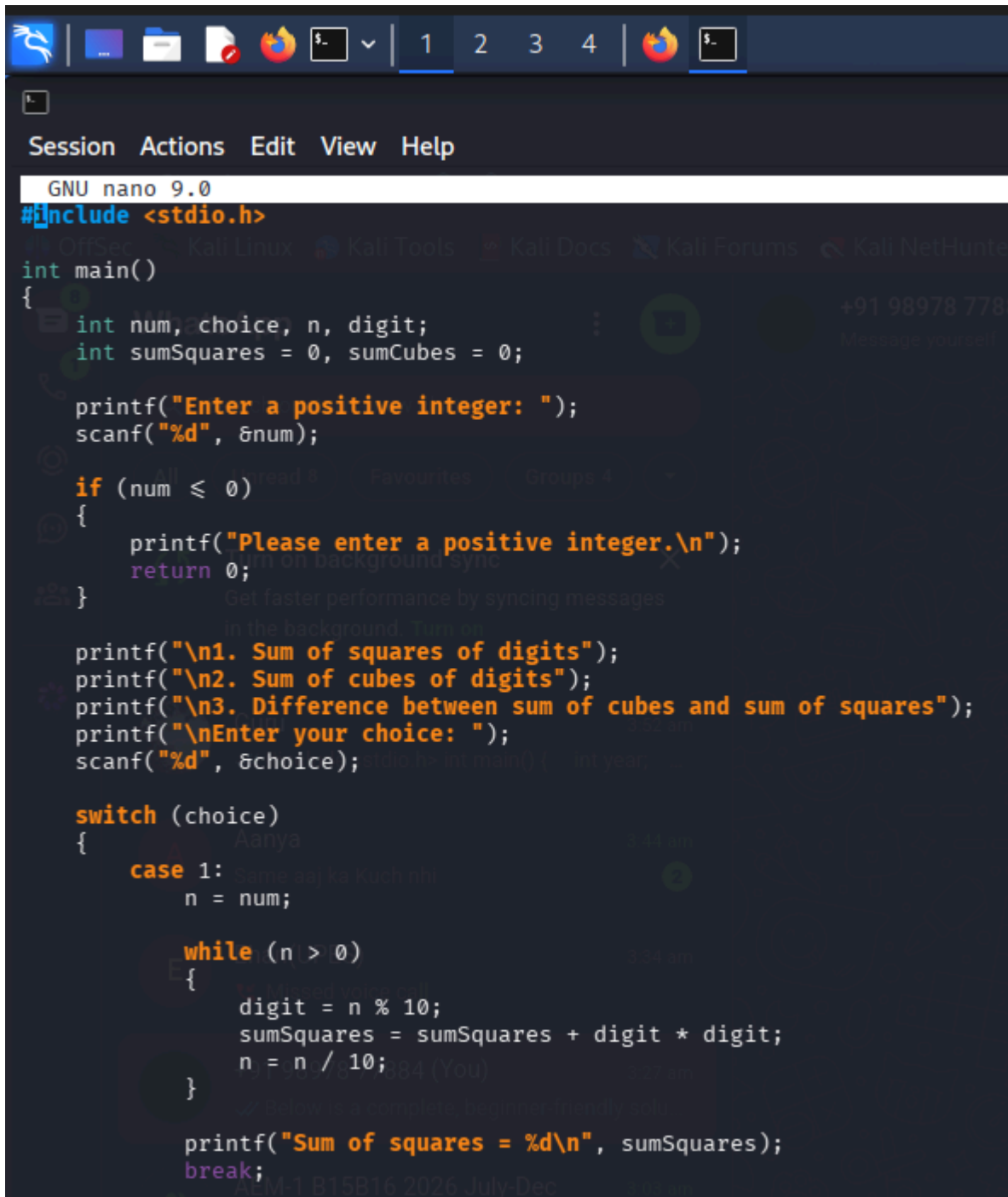
Sum of cubes:

$$\begin{aligned}1^3 + 2^3 + 3^3 \\&= 1 + 8 + 27 \\&= 36\end{aligned}$$

Difference:

$$36 - 14 = 22$$

3. Program, Compilation and Execution:



The screenshot shows a Kali Linux desktop environment. At the top, there is a taskbar with icons for a web browser, file manager, terminal, and other applications. Below the taskbar, a window titled "GNU nano 9.0" is open, displaying a C program. The program is designed to calculate the sum of squares and sum of cubes of the digits of a positive integer. It includes a menu bar with "Session", "Actions", "Edit", "View", and "Help". The code is as follows:

```
GNU nano 9.0
#include <stdio.h>
int main()
{
    int num, choice, n, digit;
    int sumSquares = 0, sumCubes = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    if (num <= 0)
    {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("\n1. Sum of squares of digits");
    printf("\n2. Sum of cubes of digits");
    printf("\n3. Difference between sum of cubes and sum of squares");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            n = num;

            while (n > 0)
            {
                digit = n % 10;
                sumSquares = sumSquares + digit * digit;
                n = n / 10;
            }

            printf("Sum of squares = %d\n", sumSquares);
            break;
```

```

case 2:
    for (n = num; n > 0; n = n / 10)
    {
        digit = n % 10;
        sumCubes = sumCubes + digit * digit * digit;
    }

    printf("Sum of cubes = %d\n", sumCubes);
    break;

case 3:
    n = num;
    do
    {
        digit = n % 10;
        sumSquares = sumSquares + digit * digit;
        sumCubes = sumCubes + digit * digit * digit;
        n = n / 10;
    } while (n > 0);

    printf("Difference = %d\n",
           sumCubes - sumSquares);
    break;

default:
    printf("Invalid choice.\n");
}

return 0;
}

```

